



SDI/TDI Gas Laws Equations Worksheet

The following abbreviations appear within the formulas on this document:

- "P" for Pressure
- "D" for Depth
- "T" for Temperature
- "V" for Volume
- "°C" for Degrees Celsius (or Centigrade)
- "°F" for Degrees Fahrenheit
- Subscript "1" for a Starting Value
- Subscript "2" for an Ending Value
- "msw" for Metres of Seawater
- "mfw" for Metres of Freshwater
- "fsw" for Feet of Seawater
- "ffw" for Feet of Freshwater

Buoyancy Calculations

Buoyancy Characteristics of an Object

Metric:

Buoyancy = weight of object - [(litres displaced) x (kg per litre of water)]

Imperial:

Buoyancy = weight of object - [(cu ft displaced) x (lbs per cu ft of water)]

Offsetting Negative Buoyancy

Metric:

litres required = [kg of negative buoyancy] / [kg per litre of water]

Imperial:

cu ft required = [lbs of negative buoyancy] / [lbs per cu ft of water]

Pressure and Depth Conversions

Gauge Pressure

psi to bar:

bar = psi / 14.7

bar to psi:

psi = bar x 14.7

Depth

Imperial to Metric:

metres = feet / 3.3

Metric to Imperial:

feet = metres x 3.3

Depth-to-Pressure (Seawater)

Metric:

$P = [D / 10] + 1$ or $P = [D + 10] / 10$

Imperial:

$P = [D / 33] + 1$ or $P = [D + 33] / 33$

Depth-to-Pressure (Freshwater)

Metric:

$P = [D / 10.3] + 1$ or $P = [D + 10.3] / 10.3$

Imperial:

$P = [D / 34] + 1$ or $P = [D + 34] / 34$

Pressure-to-Depth (Freshwater)

Metric:

$D = [P - 1] \times 10.3$ or $D = [P \times 10.3] - 10.3$

Imperial:

$D = [P - 1] \times 34$ or $D = [P \times 34] - 34$

Depth Seawater-to-Freshwater

Metric:

$mfw = msw \times 1.03$

Imperial:

$ffw = fsw \times 1.03$



Pressure and Depth Conversions Continued

Pressure-to-Depth (Seawater)

Metric:

$$D = [P - 1] \times 10 \text{ or } D = [P \times 10] - 10$$

Imperial:

$$D = [P - 1] \times 33 \text{ or } D = [P \times 33] - 33$$

Depth Freshwater-to-Seawater

Metric:

$$\text{msw} = \text{mfw} / 1.03$$

Imperial:

$$\text{fsw} = \text{ffw} / 1.03$$

Pressure and Volume Calculations

Calculating Volume Following Change in Pressure

Metric and Imperial:

$$V_2 = [P_1 \times V_1] / P_2$$

Calculating Pressure Following Change in Volume

Metric and Imperial:

$$P_2 = [P_1 \times V_1] / V_2$$

Calculating Gas Consumption Time Based on Known Consumption at a Specific Depth

Metric and Imperial:

$$\text{minutes}_2 = [P_1 \times \text{minutes}_1] / P_2$$

Calculating Depth at which a Gas Supply will Last Based on Known Consumption at a Specific Depth

Metric and Imperial:

$$P_2 = [P_1 \times \text{minutes}_1] / \text{minutes}_2$$

Calculating Relative Change in Ambient Pressure

Metric and Imperial:

$$\text{relative change} = P_2 / P_1$$

Temperature, Volume and Pressure Calculations

Calculating Volume Following Change in Temperature

Metric:

$$V_2 = (C_2 + 273) \times [V_1 / (C_1 + 273)]$$

Imperial:

$$V_2 = (F_2 + 460) \times [V_1 / (F_1 + 460)]$$

Calculating Pressure Following Change in Temperature

Metric:

$$\text{bar}_2 = [(C_2 + 273) \times [(\text{bar}_1 + 1) / (C_1 + 273)]] - 1$$

Imperial:

$$\text{psi}_2 = [(F_2 + 460) \times [(\text{psi}_1 + 14.7) / (F_1 + 460)]] - 14.7$$

Quick Estimates for Temperature and Pressure

Metric:

$$1^\circ\text{C} = 0.6 \text{ bar}$$

Imperial:

$$1^\circ\text{F} = 5 \text{ psi}$$

Combined Gas Law

Metric and Imperial:

$$(P_1 \times V_1) / T_1 = (P_2 \times V_2) / T_2$$

Ideal Gas Law

Metric and Imperial:

$$P \times V = (n \times R) \times T \text{ or } PV = nRT$$



Partial Pressure Calculations

Calculating PO₂ and PN₂

Metric and Imperial:

$$PO_2 = P \times FO_2 \text{ and } PN_2 = P \times FN_2$$

Calculating FO₂ and FN₂

Metric and Imperial:

$$FO_2 = PO_2 / P \text{ and } FN_2 = PN_2 / P$$

Calculating Max Depth of a Nitrox Mixture that does not Exceed a Specified PO₂

Metric and Imperial:

$$P = PO_2 / FO_2$$

Equivalent Air Depth

Metric:

$$EAD = [(FN_2 / 0.79) \times (D + 10)] - 10$$

Imperial:

$$EAD = [(FN_2 / 0.79) \times (D + 33)] - 33$$

GROUP DESIGNATION

* Highest repetitive group that can be achieved at this depth regardless of bottom time.

Depth (feet)	No-Deco Limits (min)	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	Z
10	Unlimited	57	101	158	245	426	*										
15	Unlimited	36	60	88	121	163	217	297	449	*							
20	Unlimited	26	43	61	82	106	133	165	205	256	330	461	*				
25	1102	20	33	47	62	78	97	117	140	166	198	236	285	354	469	992	1102
30	371	17	27	38	50	62	76	91	107	125	145	167	193	223	260	307	371
35	232	14	23	32	42	52	63	74	87	100	115	131	148	168	190	215	232
40	163	12	20	27	36	44	53	63	73	84	95	108	121	135	151	163	
45	125	11	17	24	31	39	46	55	63	72	82	92	102	114	125		
50	92	9	15	21	28	34	41	48	56	63	71	80	89	92			
55	74	8	14	19	25	31	37	43	50	56	63	71	74				
60	63	7	12	17	22	28	33	39	45	51	57	63					
70	48	6	10	14	19	23	28	32	37	42	47	48					
80	39	5	9	12	16	20	24	28	32	36	39						
90	33	4	7	11	14	17	21	24	28	31	33						
100	25	4	6	9	12	15	18	21	25								
110	20	3	6	8	11	14	16	19	20								
120	15	3	5	7	10	12	15										
130	12	2	4	6	9	11	12										
140	10	2	4	6	8	10											
150	8		3	5	7	8											
160	7		3	5	6	7											
170	6			4	6												
180	6			4	5	6											
190	5			3	5												

WARNING:

Even strict compliance with these charts will not guarantee avoidance of decompression sickness. Conservative usage is strongly recommended.

* Dives following surface intervals longer than this are not repetitive dives. Use actual bottom times in the Air Decompression Tables to compute decompression for such dives.

Locate the diver's repetitive group designation from his previous dive along the diagonal line above the table. Read horizontally to the interval in which the diver's surface interval lies.

Next, read vertically downward to the new repetitive group designation. Continue downward in this same column to the row that represents the depth of the repetitive dive. The time given at the intersection is residual nitrogen time, in minutes, to be applied to the repetitive dive.

Repetitive Dive Depth (feet)	Z	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
10	**	**	**	**	**	**	**	**	**	**	**	**	427	246	159	101
15	**	**	**	**	**	**	**	**	**	450	298	218	164	122	89	61
20	**	**	**	**	**	**	**	**	**	462	331	257	206	166	134	106
25	†	†	470	354	286	237	198	167	141	118	98	79	63	48	34	21
30	372	308	261	224	194	168	146	126	108	92	77	63	51	39	28	18
35	245	216	191	169	149	132	116	101	88	75	64	53	43	33	24	15
40	188	169	152	136	122	109	97	85	74	64	55	45	37	29	21	13
45	154	140	127	115	104	93	83	73	64	56	48	40	32	25	18	12
50	131	120	109	99	90	81	73	65	57	49	42	35	29	23	17	11
55	114	105	96	88	80	72	65	58	51	44	38	32	26	20	15	10
60	101	93	86	79	72	65	58	52	46	40	35	29	24	19	14	9
70	83	77	71	65	59	54	49	44	39	34	29	25	20	16	12	8
80	70	65	60	55	51	46	42	38	33	29	25	22	18	14	10	7
90	61	57	52	48	44	41	37	33	29	26	22	19	16	12	9	6
100	54	50	47	43	40	36	33	30	26	23	20	17	14	11	8	5
110	48	45	42	39	36	33	30	27	24	21	18	16	13	10	8	5
120	44	41	38	35	32	30	27	24	22	19	17	14	12	9	7	5
130	40	37	35	32	30	27	25	22	20	18	15	13	11	9	6	4
140	37	34	32	30	27	25	23	21	19	16	14	12	10	8	6	4
150	34	32	30	28	26	23	21	19	17	15	13	11	9	8	6	4
160	32	30	28	26	24	22	20	18	16	14	13	11	9	7	5	4
170	30	28	26	24	22	21	19	17	15	14	12	10	8	7	5	3
180	28	26	25	23	21	19	18	16	14	13	11	10	8	6	5	3
190	26	25	23	22	20	18	17	15	14	12	11	9	8	6	5	3

Residual Nitrogen Times (Minutes)

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** Residual Nitrogen Time cannot be determined using this table.

† Read vertically downward to the 30 fsw repetitive dive depth. Use the corresponding residual nitrogen times to compute the equivalent single dive time. Decompress using the 30 fsw air decompression table.

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US Navy Air Decompression Tables

	Time to First Stop		DECO STOP		Repetitive Group
	Bottom Time min	minsec	30	20	
50 FSW	92	1:40	0	1:40	M
	95	1:00	2	3:40	M
	100	1:00	4	5:40	N
	110	1:00	8	9:40	O
	120	1:00	21	22:40	O
	130	1:00	34	35:40	Z
	140	1:00	45	46:40	Z
	150	1:00	56	57:40	Z
	160	1:00	78	79:40	Z
	170	1:00	96	97:40	Z
	180	1:00	111	112:40	Z
	190	1:00	125	126:40	Z
	200	1:00	136	137:40	Z

55 FSW	30		20	
	min	minsec	minsec	
	74	1:50	0	1:50 L
	75	1:10	1	2:50 L
	80	1:10	4	5:50 M
	90	1:10	10	11:50 N
	100	1:10	17	18:50 O
	110	1:10	34	35:50 O
	120	1:10	48	49:50 Z
	130	1:10	59	60:50 Z
	140	1:10	84	85:50 Z
	150	1:10	105	106:50 Z
	160	1:10	123	124:50 Z
	170	1:10	138	139:50 Z
	180	1:10	151	152:50 Z

60 FSW	30		20		
	min	minsec		minsec	
	63	2:00	0	2:00	K
	65	1:20	2	4:00	L
	70	1:20	7	9:00	L
	80	1:20	14	16:00	N
	90	1:20	23	25:00	O
	100	1:20	42	44:00	Z
	110	1:20	57	59:00	Z
	120	1:20	75	77:00	Z
	130	1:20	102	104:00	Z
	140	1:20	124	126:00	Z
150	1:20	143	145:00	Z	
160	1:20	158	160:00	Z	

70 FSW	30		20	
	min	minsec	minsec	
	48	2:20	0	2:20 K
	50	1:40	2	4:20 K
	55	1:40	9	11:20 L
	60	1:40	14	16:20 M
	70	1:40	24	26:20 N
	80	1:40	44	46:20 O
	90	1:40	64	66:20 Z
	100	1:40	88	90:20 Z
110	1:40	120	122:20 Z	
120	1:40	145	147:20 Z	
130	1:40	167	169:20 Z	

80 FSW	30		20		
	min	minsec	minsec		
	39	2:40	0	2:40	J
	40	2:00	1	3:40	J
	45	2:00	10	12:40	K
	50	2:00	17	19:40	M
	55	2:00	24	26:40	M
	60	2:00	30	32:40	N
	70	2:00	54	56:40	O
	80	2:00	77	79:40	Z
90	2:00	114	116:40	Z	
100	1:40	1	147	150:20	Z
110	1:40	6	171	179:20	Z

	Time to First Stop		DECO STOP		Repetitive Group	
	Bottom Time		40	FSW	20	Total Deco Time
	min	minsec				minsec
90 FSW	33	3:00	0	3:00	J	
	35	2:20	4	7:00	J	
	40	2:20	14	17:00	L	
	45	2:20	23	26:00	M	
	50	2:20	31	34:00	N	
	55	2:20	39	42:00	O	
	60	2:20	56	59:00	O	
	70	2:20	83	86:00	Z	
	80	2:00	5	125	132:40	Z
	90	2:00	13	158	173:40	Z

	40		30		20	
	min	minsec	min	minsec	min	minsec
100 FSW	25	3:20	0	3:20	H	
	30	2:40	3	6:20	J	
	35	2:40	15	18:20	L	
	40	2:40	26	29:20	M	
	45	2:40	36	39:20	N	
	50	2:40	47	50:20	O	
	55	2:40	65	68:20	Z	
	60	2:40	81	84:20	Z	
	70	2:20	11	124	138:00	Z
	80	2:20	21	160	184:00	Z

	40		30		20	
	min	minsec	min	minsec	min	minsec
110 FSW	20	3:40	0	3:40	H	
	25	3:00	5	8:40	I	
	30	3:00	14	17:40	K	
	35	3:00	27	30:40	M	
	40	3:00	39	42:40	N	
	45	3:00	50	53:40	O	
	50	3:00	71	74:40	Z	
	55	2:40	5	85	93:20	Z
	60	2:40	13	111	127:20	Z
	70	2:40	26	155	184:20	Z

	40		30		20	
	min	minsec	min	minsec	min	minsec
120 FSW	15	4:00	0	4:00	F	
	20	3:20	4	8:00	H	
	25	3:20	9	13:00	J	
	30	3:20	24	28:00	L	
	35	3:20	38	42:00	N	
	40	3:00	2	49	54:40	O
	45	3:00	3	71	77:40	Z
	50	3:00	10	85	98:40	Z
	55	3:00	19	116	138:40	Z
	60	3:00	27	142	172:40	Z

130 FSW	40		30		20	
	min	min:sec	min:sec			
	12	4:20			0	4:20 F
	15	3:40			3	7:20 G
	20	3:40			8	12:20 I
	25	3:40			17	21:20 K
	30	3:20	2	32	38:00 M	
	35	3:20	5	44	53:00 O	
	40	3:20	6	66	76:00 Z	
	45	3:00	1	11	84	99:40 Z
50	3:00	2	20	118	143:40 Z	
55	3:00	4	28	146	181:40 Z	
60	3:00	12	28	170	213:40 Z	

140 FSW	Time to First Stop		DECO STOP				Repetitive Group	
	Bottom Time		50	40	30	20	Total Deco Time	
	min	min:sec					min:sec	
	10	4:40				0	4:40	E
	15	4:00				5	9:40	H
	20	4:00				13	17:40	J
	25	3:40			3	24	31:20	L
	30	3:40			7	37	48:20	N
	35	3:20		2	7	58	71:00	O
	40	3:20		4	7	82	97:00	Z
45	3:20		5	18	114	141:00	Z	
50	3:20		8	27	145	184:00	Z	
55	3:00	1	15	29	171	219:40	Z	

	50		40		30		20	
	min	minsec	min	minsec	min	minsec	min	minsec
150 FSW	8	5:00	0	5:00	E			
	10	4:20	2	7:00	F			
	15	4:20	8	13:00	H			
	20	4:00	2	15	21:40	K		
	25	4:00	7	29	40:40	M		
	30	3:40	4	7	45	60:20	O	
	35	3:40	6	7	74	91:20	Z	
	40	3:20	2	6	14	106	132:00	Z
	45	3:20	3	8	24	142	181:00	Z
	50	3:20	4	14	28	170	220:00	Z

	60		50		40		30		20	
	min	minsec	min	minsec	min	minsec	min	minsec	min	minsec
160 FSW	7	5:20	0	5:20	E					
	10	4:40	4	9:20	F					
	15	4:20	2	10	17:00	I				
	20	4:00	1	4	19	28:40	L			
	25	4:00	4	7	35	50:40	N			
	30	3:40	2	6	7	62	81:20	Z		
	35	3:40	4	6	8	89	111:20	Z		
	40	3:40	6	6	21	134	171:20	Z		
	45	3:20	2	5	11	28	166	216:00	Z	

	60		50		40		30		20	
	min	minsec	min	minsec	min	minsec	min	minsec	min	minsec
170 FSW	6	5:40	0	5:40	D					
	10	5:00	6	11:40	G					
	15	4:40	3	13	21:20	J				
	20	4:20	3	6	24	38:00	M			
	25	4:00	1	7	7	41	60:40	O		
	30	4:00	5	7	7	77	100:40	Z		
	35	3:40	2	6	6	15	120	153:20	Z	
	40	3:40	4	6	9	25	158	206:20	Z	
	45	3:40	5	7	16	28	197	257:20	Z	

	70		60		50		40		30		20	
	min	minsec	min	minsec	min	minsec	min	minsec	min	minsec	min	minsec
180 FSW	6	6:00	0	6:00	E							
	10	5:20	8	14:00	G							
	15	4:40	2	3	14	24:20	K					
	20	4:20	1	5	7	29	47:00	M				
	25	4:20	5	6	7	57	80:00	O				
	30	4:00	3	6	6	7	95	121:40	Z			
	35	3:40	1	5	6	6	22	144	188:20	Z		</

Imperial

Equivalent Air Depths, Partial Pressures of Oxygen and Maximum Operating Depths

EAD	FO ₂	21%	22%	23%	24%	25%	26%	27%	28%	29%	30%	31%	32%	33%	34%	35%	36%	37%	38%	39%	40%
30		30 0.4	30 0.5	31 0.5	32 0.5	33 0.5	34 0.6	35 0.6	36 0.6	37 0.7	38 0.7	39 0.7	40 0.8	41 0.8	42 0.8	43 0.9	44 0.9	46 0.9	47 1.0	48 1.0	49 1.0
40		40 0.5	40 0.5	41 0.6	42 0.6	43 0.6	44 0.6	46 0.7	47 0.7	48 0.8	49 0.8	50 0.8	51 0.9	53 0.9	54 0.9	55 1.0	57 1.0	58 1.1	60 1.1	61 1.2	63 1.2
50		50 0.6	51 0.6	52 0.6	53 0.7	54 0.7	55 0.7	56 0.8	58 0.8	59 0.9	60 0.9	62 0.9	63 1.0	64 1.0	66 1.1	67 1.1	69 1.2	71 1.2	72 1.3	74 1.3	76 1.4
60		60 0.6	61 0.7	62 0.7	63 0.7	64 0.8	66 0.8	67 0.9	69 0.9	70 1.0	71 1.0	73 1.1	75 1.1	76 1.1	78 1.2	80 1.2	81 1.3	83 1.4	85 1.4	87 1.5	89 1.5
70		70 0.7	71 0.7	72 0.8	74 0.8	75 0.9	76 0.9	78 1.0	80 1.0	81 1.1	83 1.1	84 1.2	86 1.2	88 1.3	90 1.3	92 1.4	94 1.4	96 1.5	98 1.6	100 1.6	102 1.6
80		80 0.8	81 0.8	82 0.9	84 0.9	86 0.9	87 1.0	89 1.0	91 1.1	92 1.1	94 1.2	96 1.3	98 1.3	100 1.4	102 1.4	104 1.5	106 1.6	108 1.6	110 1.7	113 1.8	
90		90 0.8	91 0.9	93 0.9	94 1.0	96 1.0	98 1.1	100 1.1	101 1.2	103 1.2	105 1.3	107 1.4	109 1.4	112 1.5	114 1.6	116 1.6	119 1.7				
100		100 0.9	101 0.9	103 1.0	105 1.1	107 1.1	108 1.2	110 1.2	112 1.3	114 1.3	117 1.4	119 1.5	121 1.5	123 1.6	127 1.7						
110		110 1.0	111 1.0	113 1.1	115 1.1	117 1.2	119 1.2	121 1.3	123 1.4	126 1.4	128 1.5	130 1.6	133 1.6								
120		120 1.0	121 1.1	123 1.1	126 1.2	128 1.3	130 1.3	132 1.4	134 1.5	137 1.5	139 1.6	143 1.7									
130		130 1.1	132 1.1	134 1.2	136 1.3	138 1.3	141 1.4	143 1.5	145 1.6	148 1.6	151 1.7										
140		140 1.1	142 1.2	144 1.3	146 1.3	149 1.4	151 1.5	154 1.6	156 1.6	159 1.7											



WARNING

Susceptibility to decompression sickness and oxygen toxicity can vary from person to person, and from day to day. No dive table, computer or planning device can guarantee that — even if used correctly — you will not suffer from one or both of these problems. Therefore, the user must assume all risks associated with the use of this product. Caution is recommended.

PO₂ Maximum Operating Depths

1.4	187	177	167	159	151	144	138	132	126	121	116	111	107	102	99	95	91	88	85	82
1.5	202	192	182	173	165	157	150	143	137	132	126	121	117	112	108	104	100	97	93	90
1.6	218	207	196	187	178	170	162	156	149	143	137	132	127	122	117	113	109	105	102	99

Instructions for Use

- To determine Equivalent Air Depth (EAD) and partial pressure of oxygen (PO₂), start at the FO₂ value for the mixture used.
- Move down the FO₂ column until you find the depth value that exactly equals or just exceeds the actual dive depth.

The PO₂ for this depth will appear immediately below this number.

- To determine the EAD, move left from the actual depth value to find the Equivalent Air Depth in the far left-hand column.

- As long as the actual dive depth does not exceed the Maximum Operating Depth (MOD) shown at the bottom of the table, it is okay to use the PO₂ associated with that MOD.

- Depths shown are in feet of salt water at sea level. Fractional

depth values have been rounded down the next shallower integer. Fractional PO₂ values have been rounded upward to the next higher 0.1 bar/ata.

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Metric

Equivalent Air Depths, Partial Pressures of Oxygen and Maximum Operating Depths

EAD	FO ₂	21%	22%	23%	24%	25%	26%	27%	28%	29%	30%	31%	32%	33%	34%	35%	36%	37%	38%	39%	40%
9		9 0.4	9 0.5	9 0.5	10 0.5	10 0.6	10 0.6	10 0.6	11 0.7	11 0.7	11 0.7	11 0.8	12 0.8	12 0.8	12 0.9	13 0.9	13 0.9	13 1.0	14 1.0	14 1.1	15 1.1
12		12 0.5	12 0.5	12 0.6	13 0.6	13 0.6	13 0.7	14 0.7	14 0.7	14 0.8	15 0.8	15 0.8	15 0.9	16 0.9	16 1.0	17 1.0	17 1.0	17 1.1	18 1.1	18 1.2	18 1.2
15		15 0.6	15 0.6	16 0.6	16 0.7	16 0.7	17 0.7	17 0.8	18 0.8	18 0.8	19 0.9	19 0.9	19 1.0	20 1.0	20 1.1	21 1.1	21 1.2	21 1.2	22 1.3	22 1.3	22 1.3
18		18 0.6	18 0.7	19 0.7	19 0.8	19 0.8	20 0.9	20 0.9	21 1.0	21 1.0	22 1.1	22 1.1	23 1.1	23 1.2	24 1.2	24 1.3	25 1.3	25 1.4	26 1.4	26 1.5	26 1.5
21		21 0.7	21 0.7	22 0.8	22 0.8	23 0.8	23 0.9	24 0.9	24 1.0	24 1.0	25 1.1	26 1.1	26 1.2	27 1.2	27 1.3	28 1.3	28 1.4	28 1.5	29 1.5	30 1.6	30 1.6
24		24 0.8	24 0.8	25 0.9	25 0.9	26 1.0	27 1.0	27 1.1	28 1.1	28 1.2	29 1.2	30 1.3	30 1.3	31 1.4	31 1.4	32 1.5	32 1.5	33 1.6	34 1.6	34 1.7	34 1.7
27		27 0.8	27 0.9	28 0.9	28 1.0	29 1.0	30 1.1	30 1.1	31 1.2	31 1.2	32 1.3	33 1.3	34 1.4	34 1.4	35 1.5	36 1.5	36 1.6	37 1.7	37 1.7	38 1.8	38 1.8
30		30 0.9	30 0.9	31 1.0	32 1.0	32 1.1	33 1.1	33 1.2	34 1.2	34 1.3	35 1.4	36 1.4	37 1.5	37 1.5	38 1.6	39 1.6	39 1.7	40 1.7	41 1.8		
33		33 1.0	33 1.0	34 1.1	35 1.1	35 1.2	36 1.2	36 1.3	37 1.4	37 1.4	38 1.5	39 1.6	39 1.6	40 1.7	41 1.7	41 1.8					
36		36 1.0	36 1.1	37 1.1	37 1.2	38 1.2	39 1.3	39 1.3	40 1.4	41 1.4	41 1.5	42 1.6	43 1.6	43 1.7							
39		39 1.1	39 1.1	40 1.2	40 1.2	41 1.3	42 1.3	43 1.4	43 1.4	44 1.5	45 1.5	45 1.6									
42		42 1.1	42 1.2	43 1.3	44 1.3	44 1.4	45 1.5	46 1.5	47 1.6	47 1.6											



WARNING

Susceptibility to decompression sickness and oxygen toxicity can vary from person to person, and from day to day. No dive table, computer or planning device can guarantee that — even if used correctly — you will not suffer from one or both of these problems. Therefore, the user must assume all risks associated with the use of this product. Caution is recommended.

PO₂ Maximum Operating Depths

1.4	56	53	50	48	46	43	41	40	38	36	35	33	32	31	30	28	27	26	25	25
1.5	61	58	55	52	50	47	45	43	41	40	38	36	35	34	32	31	30	29	28	27
1.6	66	62	59	56	54	51	49	47	45	43	41	40	38	37	35	34	33	32	31	30

Instructions for Use

- To determine Equivalent Air Depth (EAD) and partial pressure of oxygen (PO₂), start at the FO₂ value for the mixture used.
- Move down the FO₂ column until you find the depth value that exactly equals or just exceeds the actual dive depth.

The PO₂ for this depth will appear immediately below this number.

- To determine the EAD, move left from the actual depth value to find the Equivalent Air Depth in the far left-hand column.

- As long as the actual dive depth does not exceed the Maximum Operating Depth (MOD) shown at the bottom of the table, it is okay to use the PO₂ associated with that MOD.

- Depths shown are in Meters of salt water at sea level. Fractional

depth values have been rounded down the next shallower integer. Fractional PO₂ values have been rounded upward to the next higher 0.1 bar/ata.

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Oxygen Exposure Time Limits (CNS "Clock")

PO ₂	Single Dive Limit 24-Hour Limit	1	2	3	4	5	10	15	20	25	30	35	40	45	50	55	60	Time in Minutes
0.6	720 720	1%	1%	1%	1%	1%	1%	1%	2%	3%	3%	4%	5%	6%	7%	8%	8%	
0.7	540 540	1%	1%	1%	1%	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	11%	
0.8	450 450	1%	1%	1%	1%	1%	2%	3%	4%	6%	7%	8%	9%	10%	11%	12%	13%	
0.9	360 360	1%	1%	1%	1%	1%	3%	4%	6%	7%	8%	10%	11%	13%	14%	15%	17%	
1.0	300 300	1%	1%	1%	1%	2%	3%	5%	7%	8%	10%	12%	13%	15%	17%	18%	20%	
1.1	240 270	1%	1%	1%	2%	2%	4%	6%	8%	10%	13%	15%	17%	19%	21%	23%	25%	
1.2	210 240	1%	1%	1%	2%	2%	5%	7%	10%	12%	14%	17%	19%	21%	24%	26%	29%	
1.3	180 210	1%	1%	2%	2%	3%	6%	8%	11%	14%	17%	19%	22%	25%	28%	31%	33%	
1.4	150 180	1%	1%	2%	3%	3%	7%	10%	13%	17%	20%	23%	27%	30%	33%	37%	40%	
1.5	120 180	1%	2%	3%	3%	4%	8%	13%	17%	21%	25%	29%	33%	38%	42%	46%	50%	
1.6	45 150	2%	4%	7%	9%	11%	22%	33%	44%	56%	67%	78%	89%	100%	—	—	—	

Instructions for Use

- This chart allows you to quickly calculate what percentage of your total oxygen exposure limits you have consumed on each dive.
- To use the chart, begin by determining the partial pressure of oxygen (PO₂) experienced at the deepest point during the dive (see reverse side).
- Find this PO₂ value in the left-hand column, and then move across horizontally until you find the column that matches the length of the dive, in minutes.
- The numbers in each box represent the percentage of the total time limits consumed during the dive. The topmost value represents the percentage of the single-dive limit consumed; the lowermost value represents the percentage of the 24-hour limit consumed.
- You can combine percentage values from different columns to determine totals that are precise to the minute. For example, for a 67-minute dive, add the percentages from the 60, 5 and 2 minute columns.
- For multiple dives separated by sufficient surface intervals, it may be possible to "recover" some of the time limits consumed on previous dives. Consult the TDI advanced Nitrox Diver course manual for more information.
- Time values in this chart are based on the NOAA Oxygen Exposure Time Limits for Working Divers. Fractional percentage values have been rounded to the nearest whole number, with a minimum value of one percent.



WARNING

Susceptibility to decompression sickness and oxygen toxicity can vary from person to person, and from day to day. No dive table, computer or planning device can guarantee that — even if used correctly — you will not suffer from one or both of these problems. Therefore, the user must assume all risks associated with the use of this product. Caution is recommended.

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